Inferring Openness to Experience from Behavioral Signals in a Virtual Reality Art Gallery

Aaditya Vardhan Narain*
SERC
IIIT-Hyderabad
Hyderabad, India
aaditya.narain@research.iiit.ac.in

Raman Saxena
SERC
IIIT-Hyderabad
Hyderabad, India
raman.saxena@iiit.ac.in

Y. Raghu Reddy
Software Engineering Research Centre
IIIT
Hyderabad, India
raghu.reddy@iiit.ac.in

Abstract

Openness to Experience is a core dimension of the Five Factor Model (FFM). It is associated with aesthetic sensitivity and cognitive exploration. Traditional assessments rely on self-report and informant-report, which are susceptible to biases such as social desirability and inaccurate self-perception. Behavioral measures offer a complementary approach, but existing methods often lack controlled yet semi-naturalistic valid settings.

In this work, we design an affectively controlled virtual reality (VR) art gallery to study and infer Openness to Experience through semi-naturalistic interaction. The experiment was conducted in two phases. To ensure cultural validity, firstly, we re-evaluate a subset of artworks from the VAPS dataset on an Indian population (N=30) and construct a stimulus set of paintings and sculptures with controlled valence, arousal, and visual complexity. Secondly, in the VR art gallery, we capture five behavioral signals from participants (N=49), spanning locomotor signals (gallery dwell time, total distance walked, average walking speed), oculomotor signals (total fixation duration), and physical interaction signals (sculpture rotations).

Results show that Indian participants rate artworks as more visually complex compared to prior VAPS ratings. We further find that total gallery dwell time is strongly associated with Openness to Experience ($\rho = 0.64$, $p < 0.001$), alongside other behavioral signals. Building on these findings, we evaluate a predictive framework using five classifiers, where Support Vector Machine (RBF kernel) achieves the best performance (Accuracy = 0.821 ± 0.11) using four behavioral features.

We conclude that affectively controlled VR environments can serve as a viable paradigm for inferring Openness to Experience from behavioral interaction signals. These findings highlight the potential for implicit personality sensing in adaptive VR systems and personalized interactive environments.

CCS Concepts

• **Human-centered computing** → **Virtual reality**; Human computer interaction (HCI); • **Applied computing** → *Psychology*.



This work is licensed under a Creative Commons Attribution 4.0 International License.
ICMI '26, Napoli, Italy
© 2026 Copyright held by the owner/author(s).
ACM ISBN 979-8-4007-2318-6/26/10
https://doi.org/10.1145/3776574.3831169

Keywords

Virtual Reality; Personality Computing; Openness to Experience; Behavioral Signals; Multimodal Interaction; Machine Learning

ACM Reference Format:
Aaditya Vardhan Narain, Raman Saxena, and Y. Raghu Reddy. 2026. Inferring Openness to Experience from Behavioral Signals in a Virtual Reality Art Gallery. In *INTERNATIONAL CONFERENCE ON MULTIMODAL INTERACTION (ICMI '26), October 05–09, 2026, Napoli, Italy*. ACM, New York, NY, USA, 10 pages. https://doi.org/10.1145/3776574.3831169

1 Introduction

Personality has been defined as a pattern of thoughts, emotions, and behaviors that remain relatively consistent across time and situations [10]. Understanding personality is important, as stable traits influence how individuals think, feel, and behave, and are predictive of real-world outcomes such as job performance, well-being, and health [4]. Traditional personality assessment relies on self-report and informant-report methods, which are susceptible to biases such as self-perception errors and social desirability, while providing limited access to observable behavioral expressions of personality [18, 26].

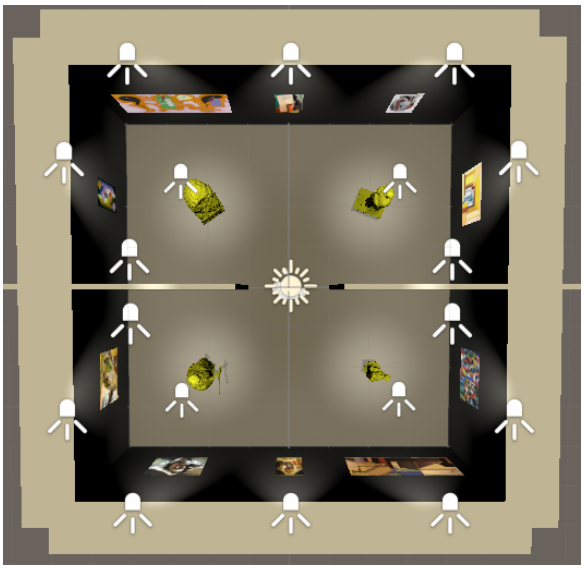


Figure 1: Top-down gallery layout with light sources marking anchor points.

To address these limitations, behavioral approaches have been proposed to infer personality from observable actions [26]. Machine learning-based personality inference has shown promise using signals such as text, social media activity, interaction logs, and digital footprints [41]. However, such data sources are often shaped by platform norms, task demands, content exposure, and other uncontrolled situational factors, making it difficult to determine whether predictive signals reflect personality-relevant behavior or artifacts of the data source.

VR offers an alternative approach by enabling the observation of behavior within controlled yet semi-naturalistic environments [7]. Compared to questionnaires or highly constrained laboratory tasks, VR allows participants to move, explore, and interact with virtual spaces while researchers retain control over environmental conditions and stimulus presentation. Prior work has explored personality expression in VR across diverse interaction paradigms, including maze navigation tasks, virtual retail environments, sports and team-coordination simulations, educational scenarios, and decision-making tasks with predefined objectives [17, 20, 32, 39, 44]. These studies demonstrate the feasibility of inferring personality from VR interaction data using behavioral, gaze-based, and physiological measures [15, 17, 37, 42]. However, behavior in such environments is often influenced by explicit task goals and performance requirements, making it difficult to distinguish personality-driven exploration from task-driven actions. Comparatively less attention has been given to theory-driven VR environments that allow free exploration and are specifically designed to elicit behaviors associated with particular personality traits.

Among the Big Five personality traits, Openness to Experience is particularly is about exploration, curiosity, and aesthetic engagement. Within the Five-Factor Model (FFM), Openness encompasses facets such as Imagination, Artistic Interests, Emotionality, Adventurousness, Intellect, and Liberalism [8, 12]. These facets reflect tendencies toward novelty seeking, intellectual curiosity, appreciation of art, and engagement with new experiences. Prior research has linked Openness to artistic interests, aesthetic appreciation, and engagement with cultural experiences, suggesting that gallery environments may provide a theoretically grounded context for observing behavioral manifestations of this trait.

Table 1: Examples of IPIP-300 [14] items related to Openness to Experience

Facet	Question
Imagination	+ Have a vivid imagination
Artistic Interests	+ Believe in the importance of art
Artistic Interests	- Do not like art
Adventurousness	+ Prefer variety to routine

Recent work has begun exploring personality within virtual art environments. Using the Open Gallery for Art Research (OGAR) platform, Rodriguez et al. examined navigation behavior and artwork engagement within a web-based virtual museum [29]. Their findings showed that individuals higher in Openness spent more time exploring the museum and viewing artworks [30]. These results provide preliminary evidence that virtual gallery environments can elicit behaviors associated with Openness. However, the environment was desktop-based rather than immersive VR, and the affective properties of artworks were not explicitly controlled.

A methodological challenge in art-based environments is that behavioral responses may be influenced not only by personality but also by the affective characteristics of the artworks themselves. Prior work has shown that emotional properties such as valence and arousal influence visual attention, gaze behavior, and exploratory patterns in virtual environments [5, 11, 19] Since personality traits are also associated with differences in affective processing [31, 33], behavioral differences within a gallery may reflect emotional responses to artworks rather than personality-related exploratory tendencies. To reduce this confound, we employ a culturally validated set of artworks with relatively neutral valence, arousal, and visual complexity.

Motivated by these considerations, we designed a VR art gallery that supports semi-naturalistic exploration of paintings and interactive sculptures while maintaining control over key stimulus characteristics. To ensure cultural validity, a subset of artworks from the VAPS dataset was re-evaluated within an Indian population and used to construct a gallery containing paintings and sculptures tailored to the target context. Prior work [22, 24, 35] has explored emotional rating within a VR environment. Participants were free to navigate the environment without explicit performance objectives, enabling the observation of naturally emerging exploratory and aesthetic behaviors associated with Openness to Experience.

Based on the theoretical framework and methodological considerations described above, the following hypotheses were formulated:

- **H1:** Spatial interaction behavior within a VR art gallery environment is significantly associated with individuals' Openness to Experience.
  - **H1a:** Total time spent in the VR gallery (gallery dwell time) is positively associated with Openness to Experience.
  - **H1b:** Spatial exploration, operationalized as total distance walked, is positively associated with Openness to Experience.
  - **H1c:** Visual engagement with artworks, operationalized as total fixation duration, is positively associated with Openness to Experience.
  - **H1d:** Physical interaction with sculptures, operationalized as average rotation interactions, is positively associated with Openness to Experience.
- **H2:** Interaction diversity, operationalized as the Shannon entropy of attention distribution across all artworks in the virtual environment, is positively associated with Openness to Experience.
- **H3:** The associations between VR interaction metrics and personality traits will be significantly stronger for Openness to Experience than for the other Big Five traits (Neuroticism, Extraversion, Agreeableness, and Conscientiousness).

This work contributes (1) a culturally grounded, affectively controlled VR stimulus dataset which is available in Table 2 and Table 3, (2) empirical evidence linking VR interaction patterns to Openness

to Experience, and (3) adoption of a behavioral prediction framework demonstrating the feasibility of inferring personality from immersive interaction data.

2 Related Work

Research investigating personality in VR spans a variety of scene designs and interaction paradigms. Existing studies include task and performance oriented environments such as team coordination and sports simulations, navigation and exploration scenarios such as mazes or virtual retail spaces, decision-making and moral-choice simulations, as well as educational and perceptual tasks [17, 20, 32, 39, 44].

Personality inference in VR relies on a diverse set of measurable signals. Behavioral metrics commonly include navigation patterns, exploration distance, interaction frequency, task performance, and decision choices [15, 17]. Visual attention is frequently captured through gaze-based features such as fixation duration, fixation frequency, and scan-path characteristics [32]. In addition, physiological signals and subjective experience measures, including presence, immersion, and cybersickness, are often incorporated to complement behavioral data [37, 43, 45]. Recent research further explores multimodal approaches that combine behavioral, perceptual, and physiological signals within machine learning frameworks for personality prediction [40, 47]. Overall, existing work demonstrates the feasibility of inferring personality traits from VR-based interaction data across diverse environments and modalities. This growing body of research highlights VR as a flexible platform for studying personality expression through observable behavior within immersive contexts.

3 Methodology

3.1 Study Overview

The study was conducted in two phases. Phase 1 focused on stimulus preparation and validation, where artworks and sculptures were rated to select an affectively neutral stimulus set suitable for the target population. Phase 2 consisted of the immersive VR experiment, in which the validated stimuli were placed in a virtual art gallery to examine relationships between behavioral metrics and personality traits.

3.2 Phase 1: Stimulus Preparation and Selection

Phase 1 aimed to construct an affectively neutral set of visual stimuli for use in the VR gallery. A subset of artworks from the VAPS dataset [9] and selected sculptures from SketchFab[1] were evaluated on emotional valence, emotional arousal, and visual complexity.

3.2.1 Neutrality Definition and Stimulus Selection. All stimuli were rated on 7-point scales. Affective neutrality was defined as mean ratings within the range of 4 ± 1 on valence and arousal. Visual complexity was constrained to the same mid-scale range to ensure perceptual comparability across stimuli, as structural properties of scenes are known to systematically influence fixation duration and the spatial distribution of attention [13]. This control minimized stimulus-driven variance in exploratory behavior. 30 artworks were selected from the VAPS dataset (rated by German audience)[9] for re-evaluation within the target population (Indian). The stimulus set comprised 14 neutral artworks forming a core baseline. An additional 12 artworks varied along a single affective dimension while keeping the remaining two within the neutral range: 4 varied in valence (two low: 2.0–3.0; two high: 5.0–6.0), 4 in arousal, and 4 in visual complexity. A further 4 artworks were randomly sampled as control stimuli. 9 sculptures were selected from Sketchfab.

3.2.2 Participants and Screening. Following the original VAPS [9] protocol, participants with formal art education were excluded. Participants who reported normal or corrected-to-normal vision were selected. They also completed the I-PANAS-SF [36] prior to the task to account for transient mood effects. The 30 paintings were divided into three groups: two groups of 13 (A and B) and a common set of 4 (C). Each participant rated paintings from either A or B, along with C, to limit session duration and reduce fatigue effects. Similarly, the 9 sculptures were divided into three groups: two groups (A and B) of 3 sculptures each, and a shared set of 3 sculptures (C).

3.2.3 VR Rating Environment and Procedure. Participants viewed each artwork from a fixed distance of approximately 2 m under uniform lighting conditions. Each stimulus was displayed for 7 s, followed by ratings of valence, arousal, and visual complexity using a 7-point scale adapted from the Self-Assessment Manikin (SAM) [6]. Presentation order was randomized across participants. Sculptures underwent an equivalent rating procedure, with a 10 s viewing duration and 3 m viewing distance to accommodate three-dimensional perception. The Phase 1 rating environment followed the same visual design principles as the gallery described in Section 3.3.1, but restricted locomotion and presented rating questions directly within the VR scene.

3.3 Phase 2: Immersive VR Art Gallery Experiment

Phase 2 used the selected stimulus set from Phase 1 within an immersive virtual art gallery. The final environment contained 14 paintings and 4 sculptures. For each experimental run, artwork locations were randomly assigned to fixed anchor points within the 8 m × 8 m gallery.

3.3.1 Apparatus and Environment. The experiment was conducted using a Meta Quest 3 headset. The gallery was designed as an 8 m × 8 m walkable space containing 14 paintings and 4 sculptures positioned at predefined anchor points, with artwork placements randomized across participants. Original painting dimensions were preserved, while SketchFab sculptures were uniformly rescaled to approximately 2 m in height and assigned a consistent material. A top-down view is shown in Figure 1. The gallery employed the Universal Render Pipeline (URP). Paintings used unlit shaders to preserve color fidelity and minimize reflections, whereas walls, floors, and other elements used lit shaders. Neutral white lighting was placed above each artwork and sculpture to ensure consistent illumination. A head-based gaze interactor recorded fixation-based behavioral metrics without explicit participant awareness. Participants could also rotate sculptures in 45° increments using controller triggers (clockwise with the left trigger and anticlockwise with the

[1] https://sketchfab.com/

right trigger), and exited the gallery using the X button on the left controller.

3.3.2 Behavioral Measures. We extracted five behavioral features from participants' interactions within the VR gallery, spanning three modalities consistent with prior VR personality research [5, 15, 32, 40]. Oculomotor engagement was captured as Total Fixation Duration (s), the cumulative duration of gaze fixations across all paintings and sculptures, recorded using a conical gaze model with a 15° aperture angle and 2 m detection distance. A fixation was defined as sustained gaze on an object exceeding 150 ms. Locomotor behavior was characterized through three signals- Total Gallery Dwell Time (s) measured as the total duration of presence within the gallery using delta time, Total Distance Walked (m) computed as the sum of Euclidean distances between consecutive timestamps, reflecting exploratory and re-visitation behavior, and Average Walking Speed ($m\,s^{-1}$) derived as the ratio of total distance walked to total gallery dwell time, where lower values indicate more attentive, artwork-focused movement. Physical interaction with sculptures was quantified as Average Sculpture Rotations, the mean number of rotation interactions performed per sculpture, capturing engagement with three-dimensional artworks.

3.3.3 Experimental Procedure. Prior to the VR session, participants completed informed consent and the IPIP-60 [23] personality questionnaire. Upon arrival, participants completed a pre-experiment Simulator Sickness Questionnaire (SSQ) [16], followed by a short tutorial designed to familiarize them with VR controls and locomotion. Participants were then instructed to freely explore the virtual gallery without specific performance goals. After exploration, participants completed the IPQ presence questionnaire [34] and a post-experiment SSQ. Presence was measured using the IPQ to verify that participants experienced sufficient immersion during the VR task. Participants were instructed to spend as long as they wanted in the gallery.

3.4 Analysis

3.4.1 ML Classification Analysis. Taking inspiration from the machine learning pipelines used in prior studies [17, 40], IPIP-60 Openness scores were dichotomized into High and Low classes using a median split (median = 3.58), producing balanced binary groups. This was done to distinguish participants with higher versus lower Openness to Experience. To identify the most informative behavioral predictors, backward sequential wrapper-based feature selection optimized with Cohen's Kappa was applied. Starting with the full feature set, the feature whose removal produced the greatest improvement in Kappa was eliminated at each iteration, generating progressively smaller subsets. To reduce selection bias and estimate generalization performance, feature selection was embedded within a nested cross-validation framework. For each outer fold, selection was performed using only the training data, after which the selected subset was used to train and evaluate the classifier on the held-out test fold. Performance was assessed using accuracy, AUC, Cohen's Kappa, sensitivity, and specificity.

3.4.2 Additional Analyses. To further examine whether observed behavioral associations were specific to Openness rather than reflecting general personality engagement, inter-correlations between behavioral measures and all Big Five traits were examined. Additionally, exploratory facet-level analyses were conducted using the six Openness facets derived from the IPIP-60.

4 Results

4.1 Phase 1: Stimulus Validation Results

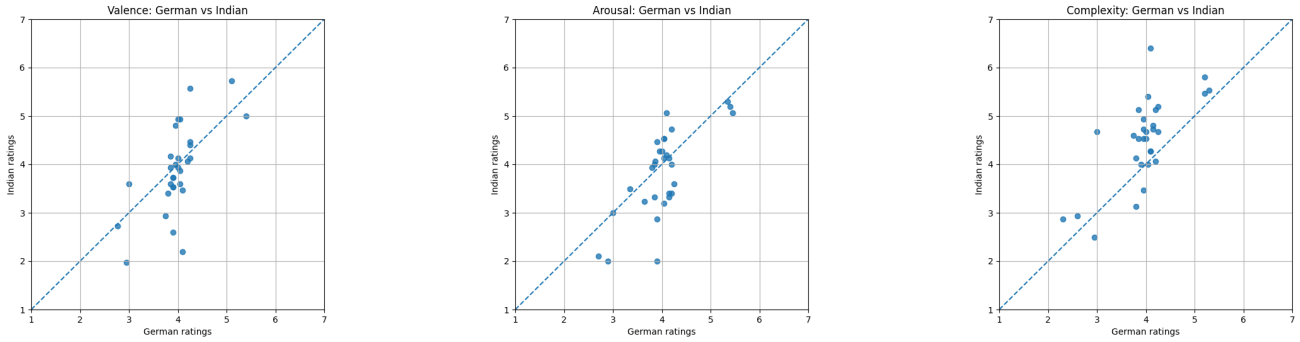
4.1.1 Participant Characteristics. 30 participants completed the stimulus validation phase. convenience sampling was employed. The sample consisted of 20 male and 10 female participants, with a mean age of 21.07 years (SD = 2.1). All participants were fluent English speakers, had no formal education in art, and reported normal or corrected-to-normal vision without color deficiencies. Participants were primarily from engineering and design backgrounds, consistent with the VAPS procedure [9] emphasizing art-novice samples. Art expertise and interest were assessed using 5 questions on a 7-point Likert scale. Reported art interest was low (M = 17.5, SD = 4.8), but slightly higher and comparable to the original VAPS sample (M = 15.93, SD = 5.41) [9]. Participants' baseline mood was evaluated using the I-PANAS-SF [36]. Positive affect (M = 16.50, SD = 2.68) was significantly higher than negative affect (M = 12.43, SD = 3.09), $t = 5.97$, $p < 0.001$, indicating an overall positive baseline mood.

4.1.2 Design and Subgroup Comparability. The rating design used 2 participant subsets that evaluated partially overlapping stimulus sets, with shared items used to assess reliability across groups. Both groups were comparable in demographic and psychological characteristics. No significant differences were observed in age ($F(1, 28) = 2.002$, $p = 0.168$), mood (positive affect: $F(1, 28) = 0.369$, $p = 0.5484$; negative affect: $F(1, 28) = 0.778$, $p = 0.3852$), or art interest ($F(1, 28) = 0.012$, $p = 0.9122$).

4.1.3 Inter-group Reliability. Reliability between participant subsets was evaluated using the shared stimuli (four paintings and 3 sculptures). Spearman correlations between group-averaged ratings showed strong agreement for visual complexity ($\rho = .955$, $p = .0008$) and arousal ($\rho = .964$, $p = .0005$), while valence showed lower, non-significant association ($\rho = .643$, $p = .1194$). Intraclass correlation coefficients (ICC) were computed using a two-way random-effects model with absolute agreement for average ratings. Reliability was excellent for visual complexity (ICC = .960, 95% CI [.79, .99], $F = 43.03$, $p < .001$) and arousal (ICC = .973, 95% CI [.87, 1.00], $F = 81.42$, $p < .001$), and good-to-excellent for valence (ICC = .907, 95% CI [.40, .98], $F = 33.73$, $p < .001$). Analyses of variance further confirmed no significant differences between groups across rating dimensions (complexity: $F = 0.009$, $p = .927$; arousal: $F = 0.041$, $p = .842$; valence: $F = 0.334$, $p = .574$), supporting pooling of ratings for subsequent analyses.

4.1.4 Comparison Between VAPS and Indian Ratings. Ratings obtained from the Indian participant sample were compared with the original VAPS ratings across valence, arousal, and complexity.

For **valence**, mean VAPS ratings (M = 3.98) and Indian ratings (M = 3.89) did not significantly differ ($t = -0.701$, $p = 0.4887$). A significant positive Spearman correlation was observed ($\rho = 0.637$, $p = 0.0002$), indicating consistency in relative ordering. For **arousal**, mean VAPS ratings (M = 4.02) and Indian ratings (M = 3.83) also



(a) Valence **(b) Arousal** **(c) Complexity**

Figure 2: Comparison between VAPS (German) ratings and Indian ratings across (a) Valence, (b) Arousal, and (c) Complexity.

Table 2: Artwork Valence (V), Arousal (A), and Complexity (C) Ratings for the original German (G) [9] and Indian (I) samples

File number	Name of artist	Title	V G	V I	A G	A I	C G	C I
10109	Raphael	Sistine Madonna	4.00	4.93	3.00	3.00	4.10	4.27
10140	Tintoretto, Jacopo	Susanna and the Elders	3.85	3.93	4.10	4.20	5.20	5.47
10323	Ingres, Jean-Auguste-Dominique	Homer's Apotheosis	4.25	4.47	4.05	4.53	5.30	5.53
10612	Cézanne, Paul	The Large Bathers	3.90	3.53	3.85	4.00	3.75	4.60
11111	Masson, André	There Is No Finished World	2.95	1.97	5.35	5.30	5.20	5.80
11120	Sedlacek, Franz	Evening Song	5.10	5.73	4.05	3.20	4.25	4.67
20402	Courbet, Gustave	Desperated Man	3.90	3.53	5.45	5.07	4.15	4.80
20620	Klimt, Gustav	Portrait of a Lady	3.00	3.60	3.85	3.33	3.90	4.00
20706	Matisse, Henri	Femme au chapeau	4.20	4.07	4.20	3.40	4.05	4.00
21011	Radiziwill, Franz	Einer von den Vielen des 20. Jahrhunderts	2.77	2.73	3.86	4.07	3.95	4.53
30212	Goyen van, Jan	A Windmill by a River	4.00	4.13	2.90	2.00	4.20	4.07
30602	Cross, Henri Edmond	Pines Along the Shore	5.40	5.00	3.90	2.00	3.80	3.13
30621	Gauguin, Paul	The Flageolet Player on the Cliff	4.25	4.13	3.95	4.27	4.15	4.73
30627	Sérusier, Paul	Seeweed Gatherer	4.25	5.57	2.70	2.10	2.60	2.93
31001	Demuth, Charles	After Sir Christopher Wren	3.95	4.80	3.90	2.87	4.10	4.27
31009	Sheeler, Charles	Western Industrial	4.25	4.40	3.35	3.50	3.80	4.13
40233	Vallayer-Coster, Anne	Still Life with Ham, Bottles and Radishes	3.85	3.60	4.15	3.33	3.95	3.47
40710	Chagall, Marc	Flowers on a Red Ground	4.10	2.20	5.40	5.20	4.25	5.20
41014	Thiebaud, Wayne	Jackpot Machine	3.90	3.73	4.15	3.40	2.30	2.87
41101	Chirico de, Giorgio	Great Metaphysical Interior	4.05	4.93	3.80	3.93	4.05	5.40
50801	Blanchard, María	Composición cubista	4.00	3.93	4.05	4.13	3.95	4.93
50803	Gleizes, Albert	On Brooklyn Bridge	3.75	2.93	4.10	5.07	4.10	6.40
50811	Duchamp, Marcel	The Passage from Virgin to Bride	4.10	3.47	3.90	4.47	4.20	5.13
50812	Bruce, Patrick Henry	Composition V	3.90	3.73	4.00	4.27	3.85	5.13
50905	Moholy-Nagy, Lazlo	A II	3.85	4.17	3.65	3.23	2.95	2.50
50932	Kupka, František	Serie CVI	3.95	4.00	4.15	4.13	3.85	4.53
50942	Vasarely, Victor	Gestalt-Zoeld	3.80	3.40	4.20	4.73	3.95	4.73
51001	Demuth, Charles	Machinery	4.05	3.87	4.25	3.60	4.00	4.53
51111	Kandinsky, Wassily	Penetrating Green	4.05	3.60	4.20	4.00	4.00	4.67
51313	Francis, Sam	Blue	3.90	2.60	4.05	4.53	3.00	4.67

showed no significant difference ($t = -1.692$, $p = 0.1014$), with a significant positive correlation ($\rho = 0.593$, $p = 0.0006$). For **complexity**, Indian ratings (M = 4.50) were significantly higher than VAPS ratings (M = 3.96), $t = 4.731$, $p = 0.0001$. Despite this shift in absolute values, a significant positive correlation was observed ($\rho = 0.641$, $p = 0.0001$), indicating consistency in relative complexity ranking across artworks.

4.1.5 The Final Selected paintings and sculptures: The virtual gallery required a final set of 14 emotionally neutral paintings and 4 emotionally neutral sculptures. The selected paintings showed mean ratings of valence = 3.88, arousal = 3.91, and complexity = 4.47. The selected sculptures showed mean ratings of valence = 4.03, arousal = 4.14, and complexity = 4.27. The final select paintings file numbers were 10109, 10612, 20402, 20620, 20706, 30621, 31009, 40233, 50801, 50812, 50932, 50942, 51001 and 51111 which can referred from Table 2. Note that these file numbers are from taken from the supplementary of VAPS dataset [9]. The selected sculptures

Table 3: Sculpture Valence (V), Arousal (A), ICMIand Complexity (C) Ratings for the Indian (I) samples

Sculpture number	Name of Sculpture	V I	A I	C I	Citation
1	Madona Sculpture	3.53	3.93	4.47	[46]
2	Indian Marble Temple Pillar	5.07	4.73	5.73	[3]
3	Walking Shakyamuni Buddha	5.27	2.33	3.57	[27]
4	Spirit of Life	5.47	4.73	5.27	[2]
5	Stone Stupa Ayuwang Temple	4.2	3.47	2.8	[21]
6	Aphrodite Kallipygos Statuette	4.23	4.5	4.57	[1]
7	Sri Chinmoy Statue	4.27	3.27	3.17	[25]
8	Harvard Yenching Institute Statue	4.07	4.87	4.87	[38]
9	Unknown Women Saint with a Book	3.93	3.6	5	[28]

were 1, 6, 7 and 8 from Table 3. While most artwork was within the neutrality range defined, 2 paintings, 20402 had a higher arousal of 5.07 and 50812 had a higher complexity of 5.13. This was due to only 12 paintings falling within the neutrality range. Do note that the 2 paintings selected outside of the range showed no deviation in Average Fixation Time per painting.

4.2 Phase 2: Immersive VR Art Gallery Experiment

4.2.1 Participants: 49 participants completed the immersive VR art gallery phase. Convenience sampling was used. The sample consisted of 34 male and 15 female participants, with a mean age of 21.55 years (SD = 3.11). The age ranged from 17 to 32 and comprised participants from three Universities. Participants were primarily from engineering and design backgrounds. Despite this gender imbalance, Mann-Whitney U tests found no significant gender differences in any personality trait or behavioral metric ($p > .34$). All participants were fluent English speakers, had no formal education in art, and reported normal or corrected-to-normal vision without color deficiencies. Participants reported a moderate level of overall presence (M = 4.75, SD = 0.62), indicating that the virtual environment was reasonably immersive. This was driven by high general presence (M = 5.45) and spatial presence (M = 5.28), suggesting that users felt a clear sense of "being there" and were able to situate themselves within the space. Participants showed an increase in simulator sickness following VR exposure, with mean SSQ scores rising from 10.53 (SD = 13.04) pre-exposure to 15.49 (SD = 19.88) post-exposure, resulting in an average change of 4.96 (SD = 15.45). A small group of participants (n=6) showed a significant increase but no participant quit the experiment due to simulator sickness and a few of these reported the effects of sickness a couple of minutes after taking off the headset rather than during the experiment.

4.2.2 H1 Results:

H1a: A Spearman correlation was conducted to examine the relationship between total gallery dwell time in the VR gallery and Openness to Experience. Results indicated a strong positive association ($\rho = 0.64$, $p < .001$), supporting H1a. This suggests that individuals higher in Openness to Experience tend to spend more time exploring the VR art gallery.

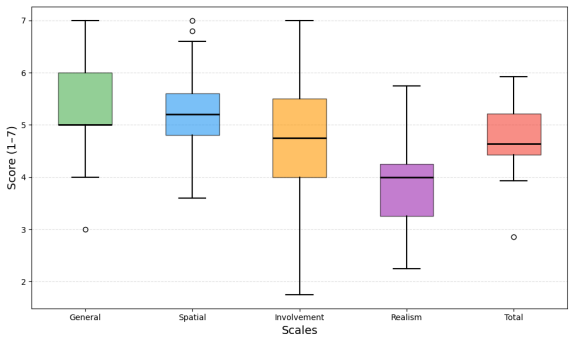


Figure 3: Presence Scores

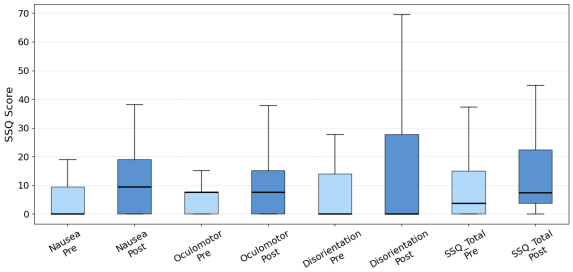


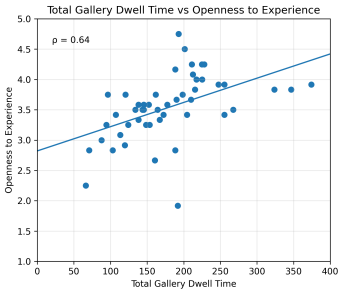
Figure 4: SSQ Scores

H1b: Spatial exploration, operationalized as total distance walked within the VR environment, showed a moderate positive correlation with Openness to Experience ($\rho = 0.37$, $p = .008$), supporting H1b. Additionally, average walking speed was negatively correlated with Openness to Experience ($\rho = -0.43$, $p < .01$), indicating that individuals higher in Openness to Experience tend to move more slowly through the environment.

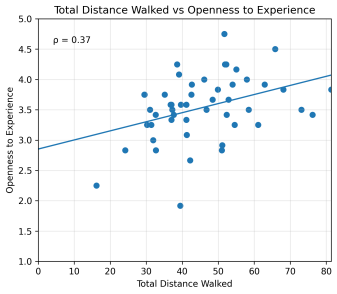
H1c: Visual engagement, operationalized as the sum of fixation durations across all artworks, showed a strong positive association with Openness to Experience ($\rho = 0.54$, $p < .001$), supporting H1c. This indicates that individuals higher in Openness to Experience tend to spend more time visually engaging with artworks.

H1d: Average Sculpture Rotations showed a weak and non- significant association with Openness to Experience ($\rho = 0.14$, $p > .05$). Thus, H1d was not supported, suggesting that interaction frequency is not meaningfully related to Openness to Experience.

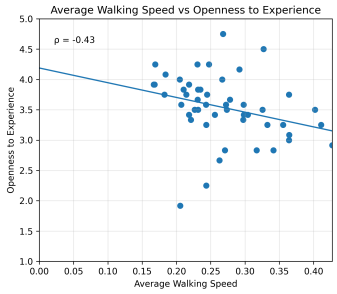
4.2.3 H2: Diversity, operationalized using Shannon entropy of attention distribution across artworks, showed a weak positive correlation with Openness to Experience ($\rho = 0.34$, $p < .05$). However, this association did not survive Bonferroni correction for multiple comparisons ($\alpha = .01$). Thus, H2 was not supported after correction. All significant results of H1 were supporterd post correction.



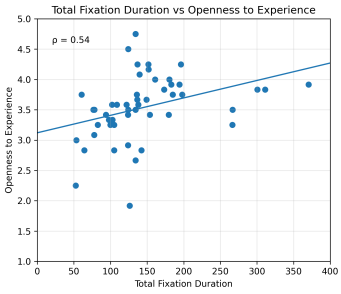
(a) Total Gallery Dwell Time



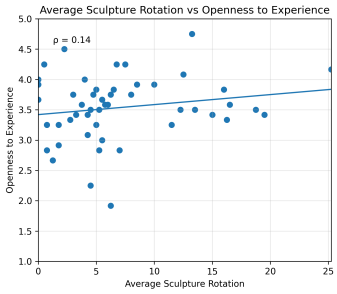
(b) Total Distance Walked



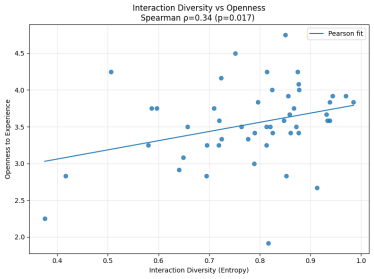
(c) Average Walking Speed



(d) Total Fixation Time



(e) Average Sculpture Rotation



(f) Entropy

Figure 5: Plots with Spearman Correlation for H1a, H1b, H1c, H1d and H2

4.2.4 H3: To assess whether behavioral metrics were more strongly associated with Openness to Experience than with other personality traits, Steiger's Z tests were conducted to compare dependent correlations. Results showed that total time spent in the gallery ($r = 0.642$) and total fixation time ($r = 0.539$) were significantly more strongly associated with Openness to Experience than with Neuroticism, Extraversion, Agreeableness, or Conscientiousness (all $p < .05$). Average movement speed ($r = -0.432$) was significantly more strongly associated with Openness to Experience compared to Neuroticism and Agreeableness, but not significantly different from Extraversion or Conscientiousness. Total distance walked ($r = 0.372$) showed a significantly stronger association with Openness to Experience only when compared to Extraversion. In contrast, rotational interaction with sculptures ($r = 0.141$) was not significantly associated with any personality trait. Overall, these findings indicate that visual engagement and spatial interaction behavioral measures are more strongly and specifically associated with Openness to Experience than movement- or interaction-based measures.

4.3 Analysis

4.3.1 ***ML Classification Results.*** Five classifiers were evaluated: Logistic Regression, Random Forest, Support Vector Machine (RBF kernel), XGBoost, and k-Nearest Neighbors (kNN). Performance was assessed using repeated stratified nested cross-validation with 5 folds repeated four times (20 outer evaluation folds). Among the evaluated models, SVM achieved the strongest performance, with an accuracy of 0.821 ± 0.11. Performance for all models and comparisons with related work are presented in Table 4. Comparisons across studies should be interpreted cautiously due to differences in environments, sample sizes, feature sets, and evaluation protocols.

Table 4: Classification performance for Openness prediction and comparison with related work

Model	Features	F	Validation Set				
			Acc	Kappa	AUC	Sensitivity	Specificity
This work - VR Art Gallery (N=49)							
Logistic Regression	Gallery Dwell Time Total Fixation Time Average Sculpture Rotation Total Distance Walked	4	$.77_{\pm.13}$	$.59_{\pm.26}$	$.77_{\pm.17}$	$.77_{\pm.20}$	$.79_{\pm.19}$
SVM (RBF)	Gallery Dwell Time Total Fixation Time Average Walking Speed Total Distance Walked	4	$\mathbf{.821_{\pm.11}}$	$\mathbf{.64_{\pm.23}}$	$\mathbf{.83_{\pm.14}}$	$\mathbf{.87_{\pm.15}}$	$.78_{\pm.18}$
XGBoost	Gallery Dwell Time Total Fixation Time Average Walking Speed Total Distance Walked Average Sculpture Rotation	5	$.73_{\pm.13}$	$.48_{\pm.26}$	$.74_{\pm.19}$	$.73_{\pm.23}$	$.74_{\pm.23}$
Random Forest (RF)	Gallery Dwell Time Total Distance Walked Total Fixation Duration	3	$.73_{\pm.12}$	$.43_{\pm.23}$	$.79_{\pm.16}$	$.62_{\pm.20}$	$\mathbf{.84_{\pm.20}}$
kNN	Gallery Dwell Time Total Distance Walked Total Fixation Duration	3	$.816_{\pm.13}$	$.64_{\pm.25}$	$.79_{\pm.17}$	$\mathbf{.87_{\pm.15}}$	$.78_{\pm.20}$
Related work							
[40] (kNN)	Eye-tracking + Behavioural	19	.70	.37	.70	.71	.69
[17] (SVM)	Behavioural (all tasks)	–	$.79_{\pm.03}$	–	–	–	–
[45] (RF)	ECG signals	–	.80	–	–	–	–

Note. **F** = number of features selected. **Acc** = Accuracy. **Kappa** = Cohen's Kappa. **AUC** = Area Under the ROC Curve.

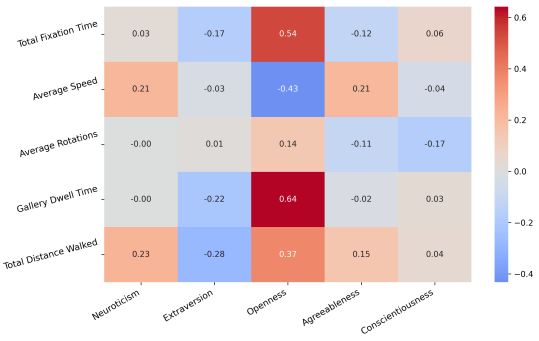


Figure 6: Spearman Correlation Heatmap between Behavioral Metrics and Personality Traits

4.3.2 Trait-Specificity Analysis. Inter-feature correlations revealed expected overlap among behavioral measures, including fixation duration and gallery dwell time (r = .865), and gallery dwell time and total distance walked (r = .752).

4.3.3 Facet Analysis. Exploratory facet-level analyses suggested that the observed associations were not driven solely by Artistic Interest. While Artistic Interest correlated with dwell time (ρ = .410) and fixation duration (ρ = .393), Imagination showed a stronger association with dwell time (ρ = .517), and Intellect was also significantly related to behavioral measures. However, these facet-level results remain exploratory, as the IPIP-60 includes only two items per facet.

5 Discussion

Openness to Experience was reflected in both visual attention and exploratory behavior within the VR gallery. Individuals higher in Openness spent more time in the gallery, exhibited longer fixation durations, explored greater distances, and moved more slowly, consistent with theories emphasizing curiosity, exploration, and aesthetic engagement. These associations were generally stronger than those observed for other Big Five traits, suggesting they reflect openness-specific tendencies rather than general VR engagement. Exploratory facet-level analyses further indicated that the effects were not limited to Artistic Interest.

Although Indian and VAPS ratings showed similar patterns for valence and arousal, Indian participants consistently rated artworks as more visually complex, underscoring the importance of validating affective stimuli within the target population rather than assuming cross-cultural equivalence.

The findings demonstrate that meaningful personality-related information can emerge from simple behavioral measures collected during unconstrained exploration, supporting the use of controlled yet naturalistic VR environments for behavioral personality computing.

Contrary to expectations, sculpture rotation frequency was unrelated, suggesting that rotational interactions may represent a relatively superficial behavior rather than a marker of aesthetic engagement.

6 Limitations

Despite the promising findings, limitations exist. The sample size (N = 49) is relatively small for machine learning, which may limit generalizability to broader populations despite the use of repeated cross-validation. The combination of a small dataset and limited feature set may increase the risk of overfitting, and model performance may not fully capture variability across unseen interaction contexts. Personality scores were dichotomized using a median split, which reduces granularity and may obscure meaningful individual differences. Additionally, the behavioral features serve as proxies for underlying psychological constructs, and richer representations such as temporal interaction patterns or physiological signals may capture these more comprehensively.

7 Future Work

We aim to expand the participant pool beyond the current sample of 49 to a larger and diverse set (N > 100) to improve the robustness and generalizability of the findings. We also plan to investigate variations in the virtual environment, including different art forms, alternative gallery layouts, and a comparative non-gallery environment, to examine whether the observed behavioral associations with Openness to Experience remain consistent across different stimulus and spatial configurations. As the present study focuses exclusively on Openness to Experience, future work will extend this framework to the remaining traits through the design of multiple VR scenarios tailored to elicit trait-relevant behaviors, with the goal of developing a generalized set of environments. Additionally, we plan to incorporate more multimodal measures, such as physiological, audio, and facial features. Finally, given the relative stability of personality traits over time, test-retest reliability will be evaluated in future studies.

8 Conclusion

This study investigated whether behavioral interaction patterns in a controlled VR art gallery are associated with Openness to Experience. To support this investigation, a culturally validated set of paintings and sculptures was first developed, revealing that Indian participants generally perceived artworks as more visually complex than reported in the original VAPS dataset. Using these validated stimuli within a freely explorable VR gallery, results showed that individuals higher in Openness spent more time in the gallery, exhibited longer fixation durations, explored greater distances, and moved more slowly, indicating stronger exploratory and aesthetic engagement. These associations were generally stronger for Openness than for the other Big Five traits. Furthermore, a machine learning framework demonstrated that Openness can be predicted from a small set of behavioral features, achieving its best performance with an SVM classifier (Accuracy = 0.821 ± 0.11). Together, these findings highlight the value of controlled yet naturalistic VR environments for studying personality expression and inferring Openness from observable behavior, as a complementary to existing questionnaire based methods.

References

[1] Ancient World 3D. 2020. 100.KZ.11 Aphrodite Kallipygos Statuette. https://sketchfab.com/3d-models/100kz11-aphrodite-kallipygos-statuette-

c01ba617ec83491195146583b70e3df9. Sketchfab asset, Creative Commons Attribution (CC BY).

[2] MCWB Architects. 2017. Spirit of Life Sculpture. https://sketchfab.com/3d-models/spirit-of-life-sculpture-91fd13978c3749f7a77f2c2719ddeb08. Sketchfab asset, Creative Commons Attribution (CC BY).

[3] Artfletch. 2023. India Marble Temple Pillar. https://sketchfab.com/3d-models/india-marble-temple-pillar-ff71713f803d4fa68d2aa6d0f5b6e4b9. Sketchfab asset, Creative Commons Attribution (CC BY).

[4] Murray R. Barrick and Michael K. Mount. 1991. THE BIG FIVE PERSONALITY DIMENSIONS AND JOB PERFORMANCE: A META-ANALYSIS. *Personnel Psychology* 44, 1 (1991), 1–26. arXiv:https://onlinelibrary.wiley.com/doi/pdf/10.1111/j.1744-6570.1991.tb00688.x doi:10.1111/j.1744-6570.1991.tb00688.x

[5] Meryem Bekler, Murat Yilmaz, and Hüseyin Emre Ilgın. 2024. Assessing feature importance in eye-tracking data within virtual reality using explainable artificial intelligence techniques. *Applied Sciences* 14, 14 (2024), 6042.

[6] Margaret M Bradley and Peter J Lang. 1994. Measuring emotion: the self-assessment manikin and the semantic differential. *Journal of behavior therapy and experimental psychiatry* 25, 1 (1994), 49–59.

[7] Pietro Cipresso and Giuseppe Riva. 2016. *Personality Assessment in Ecological Settings by Means of Virtual Reality*. John Wiley & Sons, Ltd, Chapter 18, 240–248. arXiv:https://onlinelibrary.wiley.com/doi/pdf/10.1002/9781119173489.ch18 doi:10.1002/9781119173489.ch18

[8] Paul T. Costa and Robert R. McCrae. 2008. *The revised NEO personality inventory (NEO-PI-R)*. SAGE Publications Inc., 179–198. doi:10.4135/9781849200479.n9

[9] Anna Fekete, Matthew Pelowski, Eva Specker, David Brieber, Raphaela Rosenberg, and Helmut Leder. 2023. The Vienna Art Picture System (VAPS): A data set of 999 paintings and subjective ratings for art and aesthetics research. *Psychology of Aesthetics, Creativity, and the Arts* 17, 5 (2023), 660–671. doi:10.1037/aca0000460

[10] D.C. Funder. 2019. *Personality Puzzle, 8th International Student Edition*. W. W. Norton, Incorporated. https://books.google.co.in/books?id=3SShyQEACAAJ

[11] Michal Gabay and Tom Schonberg. 2023. Passive identification of subjective preferences toward individual items using eye-tracking in a virtual reality environment. *Virtual Reality* 27, 3 (2023), 2723–2743.

[12] Lewis R Goldberg et al. 1999. A broad-bandwidth, public domain, personality inventory measuring the lower-level facets of several five-factor models. *Personality psychology in Europe* 7, 1 (1999), 7–28.

[13] John M Henderson and Taylor R Hayes. 2018. Meaning guides attention in real-world scene images: Evidence from eye movements and meaning maps. *Journal of vision* 18, 6 (2018), 10–10.

[14] International Personality Item Pool. n.d.. NEO Facets Key. https://ipip.ori.org/newNEOFacetsKey.htm. Accessed: 2026-02-28.

[15] Akrivi Katifori, Christos Lougiakis, and Maria Roussou. 2022. Exploring the effect of personality traits in VR interaction: the emergent role of perspective-taking in task performance. *Frontiers in virtual reality* 3 (2022), 860916.

[16] Robert S. Kennedy, Norman E. Lane, Kevin S. Berbaum, and Michael G. Lilienthal. 1993. Simulator Sickness Questionnaire: An Enhanced Method for Quantifying Simulator Sickness. *The International Journal of Aviation Psychology* 3, 3 (1993), 203–220. arXiv:https://doi.org/10.1207/s15327108ijap0303_3 doi:10.1207/s15327108ijap0303_3

[17] Jaikishan Khatri, Javier Marín-Morales, Masoud Moghaddasi, Jaime Guixeres, Irene Alice Chicchi Giglioli, and Mariano Alcañiz. 2022. Recognizing personality traits using consumer behavior patterns in a virtual retail store. *Frontiers in psychology* 13 (2022), 752073.

[18] Hyunji Kim, Stefano I. Di Domenico, and Brian S. Connelly. 2019. Self–Other Agreement in Personality Reports: A Meta-Analytic Comparison of Self- and Informant-Report Means. *Psychological Science* 30, 1 (2019), 129–138. arXiv:https://doi.org/10.1177/0956797618810000 doi:10.1177/0956797618810000 PMID: 30481113.

[19] Benjamin J Li, Jeremy N Bailenson, Adam Pines, Walter J Greenleaf, and Leanne M Williams. 2017. A public database of immersive VR videos with corresponding ratings of arousal, valence, and correlations between head movements and self report measures. *Frontiers in psychology* 8 (2017), 2116.

[20] Hui Liang, Zhaolin Zhang, Junjun Pan, and Jialin Fu. 2024. Assessing students' personality traits: A study of virtual reality-based educational practices. *Electronics* 13, 17 (2024), 3358.

[21] Di Luo. 2018. Stone Stupa, Ayuwang Temple. https://sketchfab.com/3d-models/stone-stupa-ayuwang-temple-afba4a79639f42d9b048ac91f34192bd. Sketchfab asset, Creative Commons Attribution (CC BY).

[22] Martin Magdin, Zoltán Balogh, Jaroslav Reichel, Jan Francisti, Štefan Koprda, and Molnár György. 2021. Automatic detection and classification of emotional states in virtual reality and standard environments (LCD): comparing valence and arousal of induced emotions. *Virtual Reality* 25, 4 (2021), 1029–1041.

[23] Jessica L. Maples-Keller, Rachel L. Williamson, Chelsea E. Sleep, Nathan T. Carter, W. Keith Campbell, and Joshua D. Miller. 2019. Using Item Response Theory to Develop a 60-Item Representation of the NEO PI–R Using the International Personality Item Pool: Development of the IPIP–NEO–60. *Journal of Personality Assessment* 101, 1 (2019), 4–15. arXiv:https://doi.org/10.1080/00223891.2017.1381968 doi:10.1080/00223891.2017.1381968 PMID: 29087223.

[24] Javier Marín-Morales, Juan Luis Higuera-Trujillo, Alberto Greco, Jaime Guixeres, Carmen Llinares, Claudio Gentili, Enzo Pasquale Scilingo, Mariano Alcañiz, and Gaetano Valenza. 2019. Real vs. immersive-virtual emotional experience: Analysis of psycho-physiological patterns in a free exploration of an art museum. *PloS one* 14, 10 (2019), e0223881.

[25] Matousekfoto. 2021. Sri Chinmoy statue. https://sketchfab.com/3d-models/sri-chinmoy-statue-4ed7a569fd684755a6c406d0ecdc4e93. Sketchfab asset, Creative Commons Attribution (CC BY).

[26] Jennifer Dodorico McDonald. 2008. Measuring personality constructs: The advantages and disadvantages of self-reports, informant reports and behavioural assessments. *Enquire* 1, 1 (2008), 1–19.

[27] Minneapolis Institute of Art. 2024. Walking Shakyamuni Buddha, 15th C CE. https://sketchfab.com/3d-models/walking-shakyamuni-buddha-15th-c-ce-14bc6412922044e996e10ae2a63dd81e. Sketchfab asset, Creative Commons Attribution (CC BY).

[28] Virtual Museums of Małopolska. 2022. Unknown woman saint with a book. https://sketchfab.com/3d-models/unknown-woman-saint-with-a-book-a2aecea000064543a8c7031504f041bc. Sketchfab asset, Creative Commons Attribution (CC BY).

[29] Rebekah M Rodriguez-Boerwinkle, Martin J Boerwinkle, and Paul J Silvia. 2023. The Open Gallery for Arts Research (OGAR): An open-source tool for studying the psychology of virtual art museum visits. *Behavior Research Methods* 55, 2 (2023), 824–842.

[30] Rebekah M Rodriguez-Boerwinkle and Paul J Silvia. 2024. Visiting virtual museums: How personality and art-related individual differences shape visitor behavior in an online virtual gallery. *Empirical Studies of the Arts* 42, 2 (2024), 439–468.

[31] Cheryl L Rusting. 1998. Personality, mood, and cognitive processing of emotional information: three conceptual frameworks. *Psychological bulletin* 124, 2 (1998), 165.

[32] Rachid Rhyad Saboundji, Kinga Bettina Faragó, and Violetta Firyaridi. 2024. Prediction of Attention Groups and Big Five Personality Traits from Gaze Features Collected from an Outlier Search Game. *Journal of imaging* 10, 10 (2024), 255.

[33] Alžběta Šašinková, Jiří Čeněk, Pavel Ugwitz, Jie-Li Tsai, Ioannis Giannopoulos, David Lacko, Zdeněk Stachoň, Jan Fitz, and Čeněk Šašinka. 2023. Exploring cross-cultural variations in visual attention patterns inside and outside national borders using immersive virtual reality. *Scientific Reports* 13, 1 (2023), 18852.

[34] Thomas Schubert, Frank Friedmann, and Holger Regenbrecht. 2001. The Igroup Presence Questionnaire (IPQ): A Measure of the Subjective Experience of Presence in Virtual Environments. *Presence: Teleoperators and Virtual Environments* 10, 3 (2001), 266–281.

[35] Luma Tabbaa, Ryan Searle, Saber Mirzaee Bafti, Md Moinul Hossain, Jittrapol Intarasisrisawat, Maxine Glancy, and Chee Siang Ang. 2022. VREED: Virtual Reality Emotion Recognition Dataset Using Eye Tracking & Physiological Measures. *Proc. ACM Interact. Mob. Wearable Ubiquitous Technol.* 5, 4, Article 178 (Dec. 2022), 20 pages. doi:10.1145/3495002

[36] Edmund R Thompson. 2007. Development and validation of an internationally reliable short-form of the positive and negative affect schedule (PANAS). *Journal of cross-cultural psychology* 38, 2 (2007), 227–242.

[37] Sebastian Oltedal Thorp, Lars Morten Rimol, and Simone Grassini. 2023. Association of the big five personality traits with training effectiveness, sense of presence, and cybersickness in virtual reality. *Multimodal Technologies and Interaction* 7, 2 (2023), 11.

[38] Alexandre Tokovinine. 2016. Harvard-Yenching Institute statue. https://sketchfab.com/3d-models/harvard-yenching-institute-statue-fc245710c4d64d0886a7f37dc522ff87. Sketchfab asset, Creative Commons Attribution (CC BY).

[39] Iris Van Sintemaartensdijk, Jan-Willem van Prooijen, Claire Nee, Marco Otte, and Paul van Lange. 2022. Personality and burglary: A virtual reality study. *Personality and Individual Differences* 196 (2022), 111712.

[40] Elena Parra Vargas, Lucia Amalia Carrasco-Ribelles, Javier Marin-Morales, Carla Ayuso Molina, and Mariano Alcañiz Raya. 2024. Feasibility of virtual reality and machine learning to assess personality traits in an organizational environment. *Frontiers in Psychology* 15 (2024), 1342018.

[41] Alessandro Vinciarelli and Gelareh Mohammadi. 2014. A Survey of Personality Computing. *IEEE Transactions on Affective Computing* 5, 3 (2014), 273–291. doi:10.1109/TAFFC.2014.2330816

[42] Zixuan Wang and Burkhard Claus Wünsche. 2024. Space War: Investigating the Impact of a Loss Aversion Strategy and Personality Traits in a VR Rowing Exergame. In *Proceedings of the 2024 Australasian Computer Science Week*. 130–139.

[43] Ari Widyanti and Hana Nadhilah Hafizhah. 2022. The influence of personality, sound, and content difficulty on virtual reality sickness. *Virtual Reality* 26, 2

(2022), 631–637.

[44] Markus Wirth, Stefan Gradl, Wolfgang A Mehringer, Richard Kulpa, Hannes Rupprecht, Dino Poimann, Annemarie F Laudanski, and Bjoern M Eskofier. 2020. Assessing personality traits of team athletes in virtual reality. In *2020 IEEE Conference on Virtual Reality and 3D User Interfaces Abstracts and Workshops (VRW)*. IEEE, 101–108.

[45] Jialan Xie, Ping Lan, Zhaonian Hu, and Guangyuan Liu. 2025. Comparative Analysis of Personality Recognition in Response to Virtual Reality and 2D Emotional Stimulus Using ECG Signals. *ACM TRANSACTIONS ON INTERACTIVE INTELLIGENT SYSTEMS* 15, 3 (2025).

[46] Jan Zachar. 2016. Madonna Sculpture. https://sketchfab.com/3d-models/madonna-sculpture-5e16f3cd478e4c8cb32dede7446a83fb. Sketchfab asset, Creative Commons Attribution (CC BY).

[47] Xintong Zhang, Di Lu, Huiqi Hu, Nan Jiang, Xianhao Yu, Jinan Xu, Yujia Peng, Qing Li, and Wenjuan Han. 2024. Personalityscanner: Exploring the validity of personality assessment based on multimodal signals in virtual reality. *arXiv preprint arXiv:2407.19728* (2024).

Safe and Responsible Statement

This work infers Openness to Experience from behavioral signals in a controlled VR art gallery, raising considerations around privacy, bias, and responsible use. All data are collected with informed consent, anonymized, and stored securely. To reduce confounding and cultural bias, we employ an affectively controlled stimulus set validated within the target population. We acknowledge that behavioral inference of personality is probabilistic and context-dependent; therefore, the system is intended strictly for research purposes and not for diagnostic or high-stakes decision-making. We emphasize transparency in modeling, limit interpretability claims, and advocate for cautious deployment to prevent misuse, profiling, or unwarranted psychological conclusions.